

Scientific Writing Lesson 3: Article/Thesis structure, Literature review, paraphrasing and tense use



Theses in your field of study and ADAM

- For today's lesson, search for and open three theses in your field of study.
- We will also look at articles, so have those ready too.
- Several of you have not yet signed and uploaded the document 'Use and declaration of digital tools and artificial intelligence on ADAM. Do that by Wednesday this week.
- Some of you uploaded your assignments in PDF format instead of in WORD format. Remove the PDFs from ADAM and upload the assignments in WORD format.

Objectives Lesson 3

- Outline the research story of your thesis / article
- Discuss tense use in research articles and theses
- Examine the study background and literature review sections of articles / theses
- Focus on some ways of reporting research and using reporting verbs
- Examine tense use in citations and paraphrasing
- Learn how to paraphrase
- Practise paraphrasing
- Do exercises on paraphrasing and formal expression

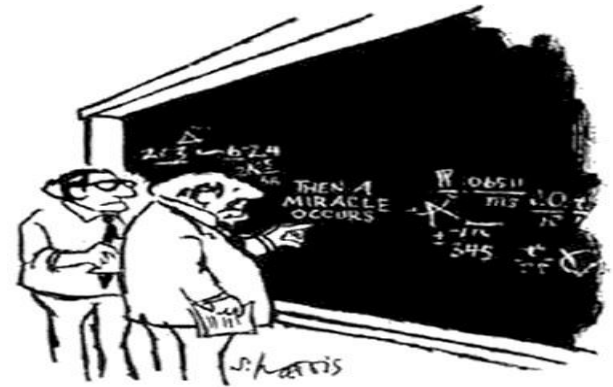
The fairy tale structure of a scientific research text...

FAIRY TALES	SCIENTIFIC PAPERS
<i>Once upon a time (background)</i>	Introduction: Why did we start?
<i>The test</i>	Methods: What did we do?
<i>The results</i>	Results: What did we find?
<i>The moral</i>	Discussion & Conclusion: What does it mean? “the message”

The message drives the story, but you can't write the message until you've finished your analysis and decided on the implications of your main finding. Since you can't start writing the story before you know it, where do we start?

Let's start at the very beginning: Structuring and outlining

Outlining creates a clear chain of arguments, linked to supporting evidence, that leads **inevitably** to your message statement.

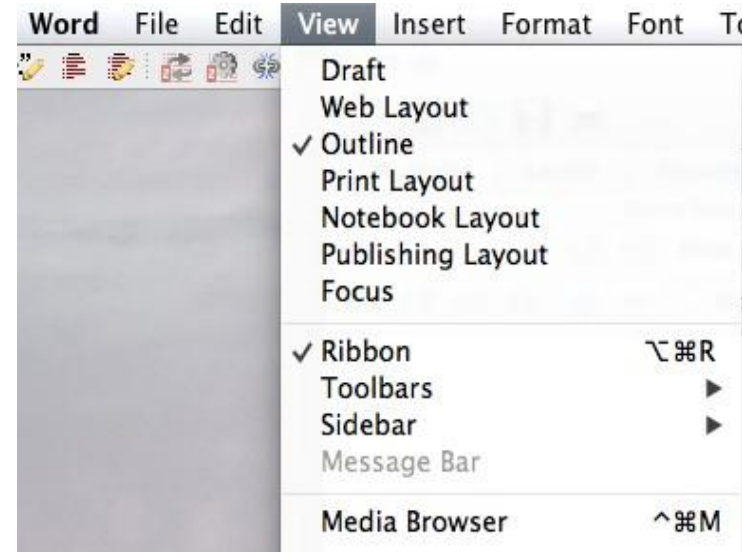


"I THINK YOU SHOULD BE MORE EXPLICIT
HERE IN STEP TWO."

[How to Make and Format an Outline in Microsoft Word](#) | Envato Tuts+ ([tutsplus.com](#))

[How To Create an Outline in Microsoft Word](#)
[\(youtube.com\)](#)

How to Use Outline View in Microsoft Word (PC & Mac)



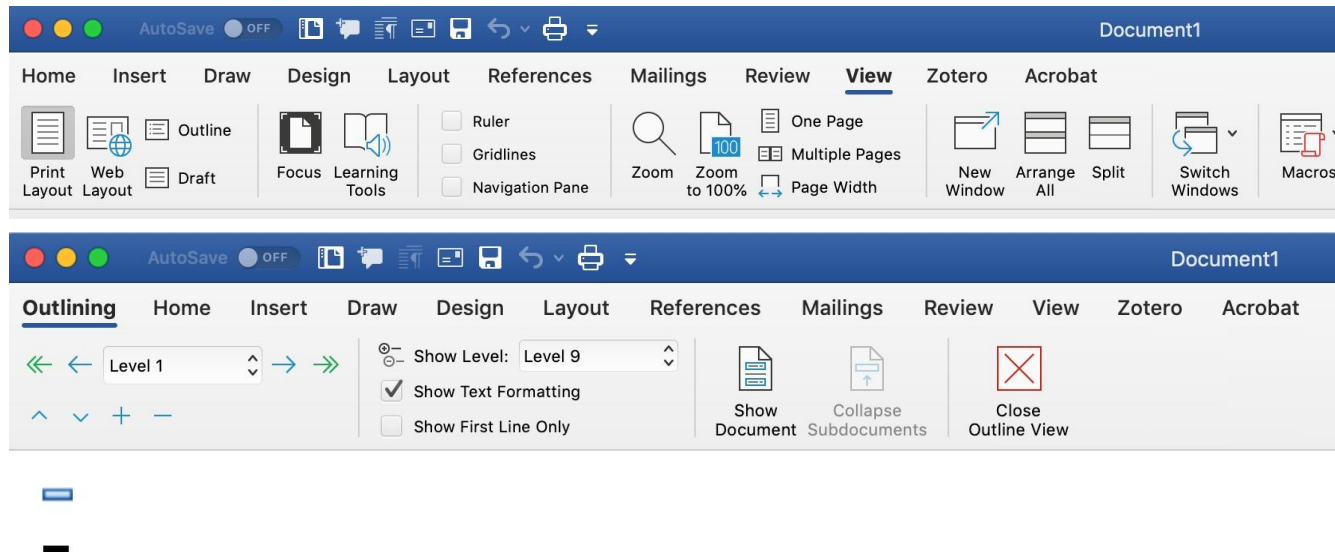
Microsoft Word is excellent for outlining. You can use outlines to take the step between mind-mapping and writing.

See the links for instructions.

Outlining differs very slightly between German and English versions of MS Word

So erstellen Sie eine Word-Gliederung in 3 Schritten:

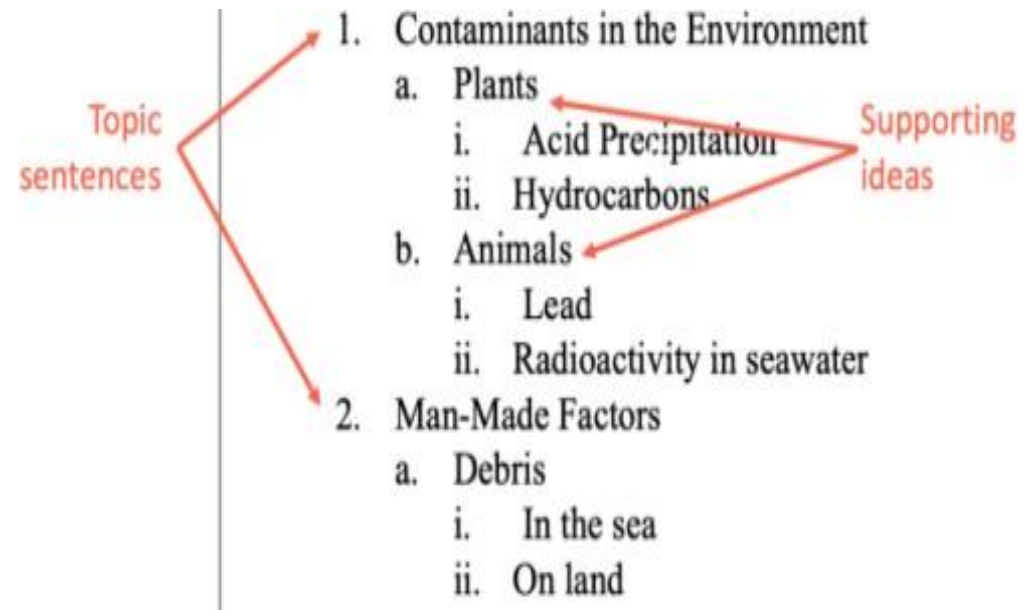
1. Öffnen Sie ein neues Word-Dokument und wählen Sie das Menü "Ansicht" aus.
2. Hier müssen Sie jetzt den Button "Gliederung" anklicken und können nun Überschriften oder Kapitelabschnitte eintragen (siehe Bild). In die verschiedenen Ebenen können Sie über die Pfeile, die Ebenen-Auswahl oder ganz einfach über die Tabulatoren-Taste (über "Shift") gelangen.
3. Wenn Sie die Gliederung vollständig eingetragen haben, müssen Sie auf "Gliederungsansicht schließen" klicken und bekommen nun Ihre Gliederung angezeigt. Das Design können Sie über die "Formatvorlagen" in "Start" noch variieren.
4. Nach Anlegen der Gliederung müssen Sie die einzelnen Punkte nur noch mit Inhalt füllen.



Fill in your outline with phrases first

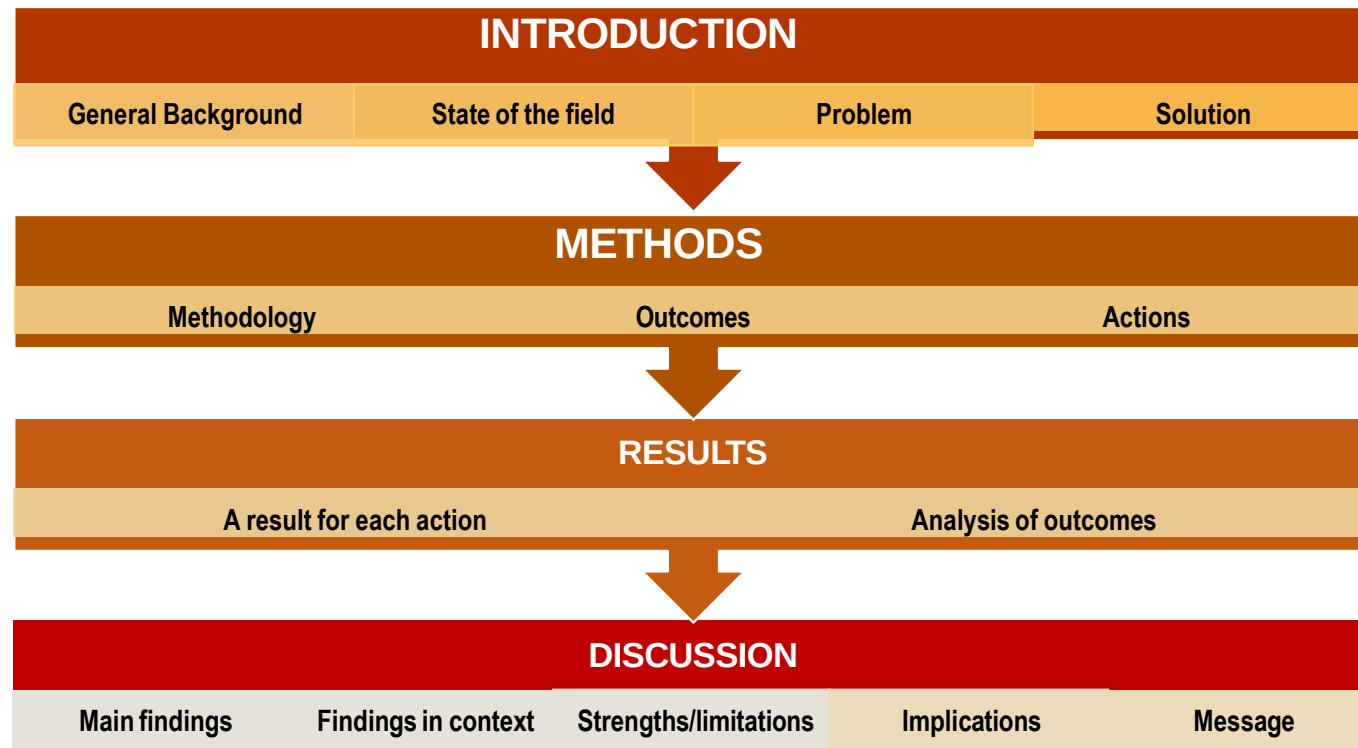
- Rewriting is hard, so jot notes first, write later
- Get your ideas and important details on paper, so you can see them all
- Focus on unfolding /“unrolling” your argument to your audience
- Don’t commit to full sentences until you think your argument unfolds properly
- Write topic sentences after the argument flows properly
- Organize detail sentences to support topic sentences

Introduction



Build a skeleton of your paper/thesis based on your outline

- Why did you start?
- What did you do?
- What did you find?
- What does it mean?



Fill in your outline as you go – it will help you to start writing

It's never too early to start writing:

- Start with your research question (problem)
- Write your study design into your manuscript
- Write your methods as you devise them
- Update your methods as you perform them
- Write your results as you obtain them
- State your limitations and strengths as you discover them
- Write the first half of your message sentence when you finish your analysis
- Write the second half of your message sentence when you think through the implications of your finding
- Put your results in context; Introduce the studies you compare your results to
- Choose your target journal and start writing paragraphs, adapting as necessary

Tense use in research articles: activity

- Which tenses are found in the Introduction section of the article? When/How are these tenses used?
- Which tense(s) is (are) used in the Methodology section?
- Which tense is mainly used in the Results section?
- Which tenses are used in the Discussion and Conclusion sections?



Examples: tense use in research articles

Abstract: Global warming **poses** great challenges, as tropical diseases now **occur** in areas where they were unknown before.

- In recent years, much research **has focused** on developing an effective malaria vaccine.
- The effect of Streptomycin on S1 **was** tested./We tested the effect / This study tested...
- Several studies **have indicated** that sea levels **are rising**.

Introduction: Streptomycin **inhibits** the growth of certain strains of Streptomyces (B). Some studies **have indicated** that... Byron et al **found** that...

Methods and Materials: Growth **was measured** by means of... We **analysed** the samples...

Results: Maximum inhibition **occurred** at pH8.2.... Table X **shows** the relationship...

Discussion and conclusion: S1 **was** most susceptible to Streptomycin at pH8.2, whereas various other Streptomyces **are** susceptible at lower pH levels (A, B, C).

Scientific Writing: a Verb Tense Review

1

Abstract

Past

To talk about actual results.

Present

To talk about general facts, the paper itself or analysis of findings.

2

Introduction

Present

To talk about general background information.

Present Perfect

To talk about previous research.

3

Methods

Past

To talk about what you did.

TIP: passive voice is common.

Present

To explain diagrams/figures.
Ex: Table 1 shows...

PAST

To talk about events that have been completed.

Ex1: We **found** that...

Ex2: Protocol X **was followed**. [Passive voice]

PRESENT PERFECT

To talk about events that started in past but are still ongoing or recently completed.

Ex: Many experiments **have focused** on...

PRESENT

To talk about events that are general fact, discuss current meanings, and suggest future applications.

Ex1: Insulin and glucagon **regulate** blood glucose levels.

Ex2: Increased hormone production **indicates**...

4

Results

Past

To talk about actual results.

Present

To explain diagrams/figures.
Ex: Table 1 shows...

5

Discussion

Present

To interpret and talk about significance of findings.

Past

To briefly summarize findings.

6

Conclusion

Past

To refer to your completed research.

Present

To talk about implications and suggest future research.



Introducing the field – providing background

1. Where does one find references to/information about the study background and other sources used?
 - The study background (at the beginning of the Introduction section)
 - The literature review (after the gap/problem statement in the Introduction)
2. What are the purposes of a study background?
 - Centrality claim to show the field is important, relevant; catch reader's attention
3. What are the purposes of a literature review?
 - Links your research to current thinking in the field of study.



The Introduction includes the study background as well as the literature review.

There are important differences between the two parts.

6 Differences between a study background and a literature review		
	Background of a study 	Literature review 
Q Where does it appear in the research paper?	In the first part of the Introduction section	After the background in the Introduction section
Q What does it present?	The context of the study	A critical analysis of existing literature
Q What is its purpose?	To highlight the significance of the study	To evaluate the progress of knowledge in the research area
Q What does it do?	Introduces the topic in context and explains how it leads up to the research problem	Helps identify the gaps in literature that led to the study
Q What does it involve?	Introducing the broad area of research and narrowing it down to the problem at hand	Analyzing the literature to show what research has been done on the topic
Q What is its length?	Is short and concise	Is lengthy and detailed
 www.editage.com Your partner in research publication     www.editage.com/insights Resources throughout your publication journey		

Differences between a study background and a literature review

- Study background is short and concise. It highlights the significance of the study; see next slide 'centrality claim'.
- Literature review usually comes after background and introduction section. It is lengthy and detailed; it evaluates the progress of knowledge in the research area.

What are the purposes of a literature review?

- Learning about available research and literature on the topic (provide a context), and incorporating it into the thesis argument
- Situating the research in a broader context
- Demonstrating what the writer knows already
- Summarizing and evaluating other people's work
- Exploring the schools of thought and 'great debates' in the area
- Identifying gaps in existing research
- Developing an understanding of theory and practice
- Interpreting, synthesizing and criticising existing research

(Murray (2002:101-108)

Literature Review task: Examine theses/articles

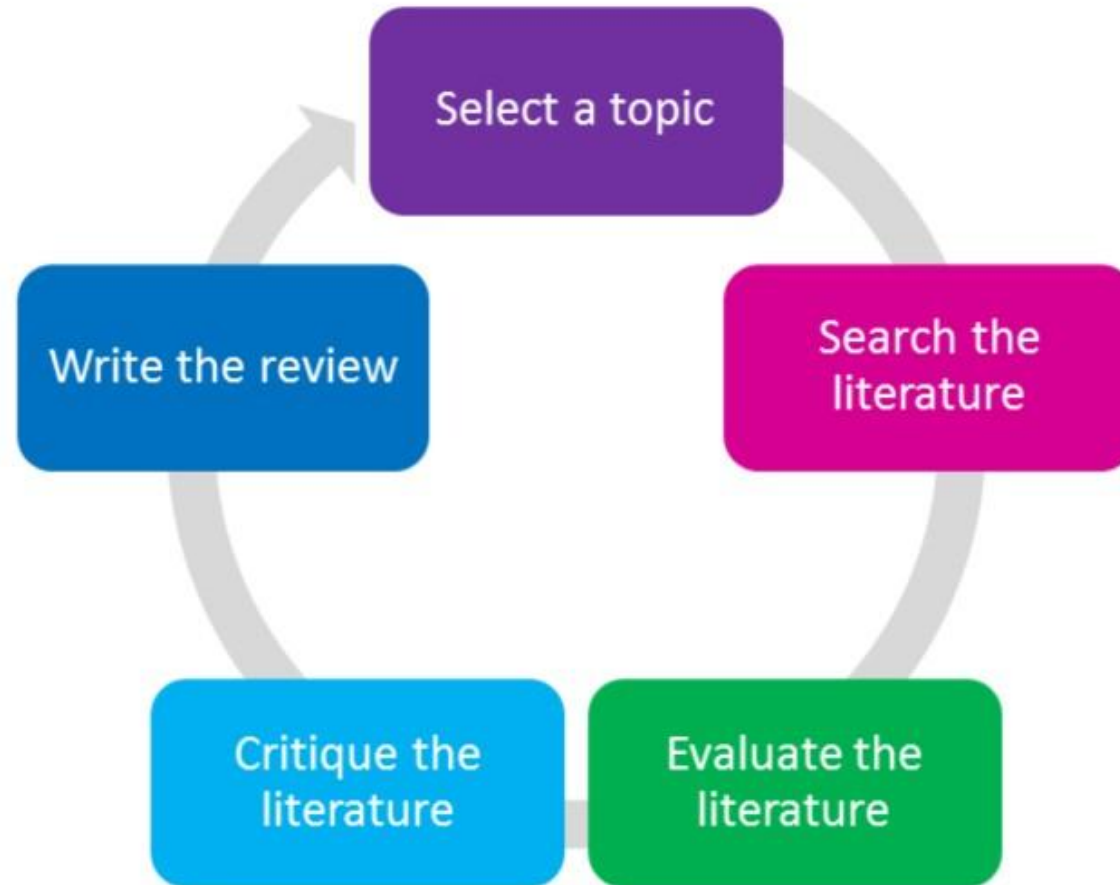
- How long is the literature review?
- Are there frequent citations? How are other works of literature in this field discussed and cited?
- What words/phrases are used to refer to such sources?
- Why is it important to refer to credible sources?
- Did the author of the thesis cite himself/herself, that is, refer to his/her own previous work? (self-reference)
- Which tenses are used when citing previous research?
- Find and write down five references to other studies, including citations, that are done in different ways.



Different kinds of literature reviews that do not form part of a paper or thesis

- **Narrative review:** gives an overview of the topic by focusing on the development of theories and concepts over time. It is often descriptive and used when only a brief review is possible. Its weakness is that it lacks the systematic methodology of more structured reviews.
- **Scoping review:** a type of research synthesis that maps the existing literature on a broad topic to identify key concepts, research gaps, and types of evidence. Unlike systematic reviews, which aim to answer specific questions, scoping reviews are exploratory and often used to assess the extent of available evidence and inform future research directions.
- **Systematic review:** characterized by its well-defined **research question** and a detailed, reproducible methodology that allows for the collection, critical appraisal, and synthesis of all relevant evidence. Inclusion and exclusion criteria are stated. Due to their structured and transparent nature, systematic reviews are particularly prevalent in fields like healthcare and social sciences, where evidence-based conclusions are needed for guiding practice and policy decisions.

Steps of a literature review



Finding the literature

<https://library.alliant.edu/screens/boolean.pdf>

Search strategies

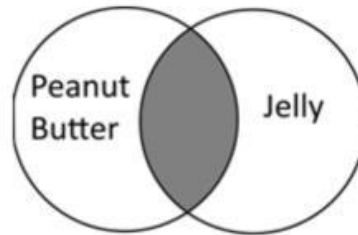
Creating search strategies combines your keywords with using advanced search options or operators that speak directly to the database

To get tips on how to create search strategies:

- Use the 'search help' on the database
- Use the specific advanced searches on the database
- **OR consult a librarian on how to use Boolean operators**

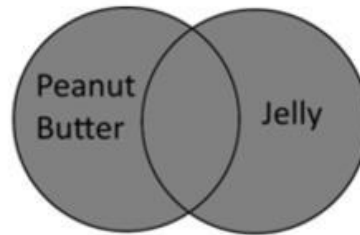
Step 2: Combine Keywords to make a search strategy

Boolean Operators - AND / OR / NOT



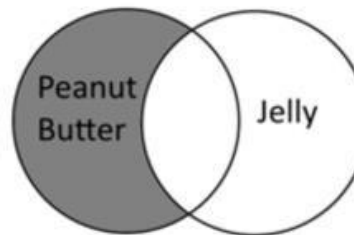
AND

Using AND, this search would only retrieve results with Peanut Butter and Jelly.



OR

Using OR, this search would retrieve results with peanut butter, with jelly, and with both.



NOT

Using NOT, this search would retrieve results with peanut butter, and exclude those with jelly or PB with jelly.

Use literature synthesis tables (see next slide and example in ‘Instructions for assignments’) to keep track of sources. These tables make it easy to compare them for your literature review and when you place your study findings in the context of other research in the Discussion and Conclusion sections.

Literature Synthesis table					
AUTHOR (S) (YEAR) TITLE	PURPOSE OF THE STUDY	METHOD / EXPERIMENT	SAMPLE / DESIGN	FINDINGS	THEMES
Mather, C. and Greenberg, S., 2003. Market Liberalisation in Post-Apartheid South Africa: the Restructuring of Citrus Exports after 'Deregulation'.	Explores the impact of market liberalisation on South Africa's citrus export Industry, specifically from a grower's perspective	Literature review conducted Qualitative data collected from interviews	Growers from 4 co-operatives within 4 specific geographical areas	Former cooperatives have faced problems in using the resources they inherited from the past, increasing differentiation between those who are able to take advantage of deregulation and those who are not. Deregulation has affected labour on farms and in packhouses but could be as a result of liberalisation of the economy having intensified structural changes in the agricultural labour market that have been present for some time.	

The literature review synthesis table

Literature synthesis table example

	Article #1	Article #2
Title	Irritable bowel syndrome subtypes: Clinical and psychological features body mass index and comorbidities	Anxiety and depression comorbidities in irritable bowel syndrome (IBS): a systematic review and meta-analysis
Source and Authors	Pubmed: Link (1) Guillaume Fond et al.	Pubmed: Link (2) Cristiane Kibune-Nagasako et al.
Key points	Three different subtypes of IBS and their frequencies in the population: IBS-Diarrhea (IBS-D), IBS-constipation (IBS-C), mixed IBS (IBS-M) Comorbidities associated with IBS: gastroesophageal reflux disease (GERD), functional dyspepsia (FD), chronic headache, fibromyalgia, obesity, anxiety and depression Anxiety and depression were found in 81.5% of the participants Which subtype associates with which comorbidities	IBS (each subtype) is associated with a high prevalence of psychological disorders, like depression and anxiety The prevalence of those mental disorders is higher in IBS patients than in control groups (general population) The treatment of those mental disorders might ease the symptoms of the IBS
Common Points	1 and 2: Use of the same scoring system for the “diagnosis” of anxiety and depression - > HAD same cut off at 8 points	1 and 2: Use of the same scoring system for the “diagnosis” of anxiety and depression -> HAD(S) same cut off at 8 points
Conclusions	Small differences were found in the prevalence of anxiety and depression in the different subtypes Generally, you can conclude from the article that a disease resulting in a comorbid state of a patient leads to higher occurrence of anxiety and depression. Psychological factors contribute to poor outcomes of the disease, influencing the severity of the IBS symptoms	Especially the comorbidities anxiety and depression should be thoroughly checked and treated. The mental state seems to influence the symptom severity and persistence of IBS patients.

Organisation in the literature review

A literature review isn't like an annotated bibliography, so you won't just go through one source at a time. Rather, organize your sources into groups as you discuss them.

These are three common ways:

1. **By time period (chronological):** Each paragraph or section addresses sources published in the same time period (e.g., one decade), and the lit review is organized from oldest to most recent findings/sources.
2. **By theme (thematic):** Each paragraph or section describes sources that focus on a specific subtopic within the overall research topic.
3. **By research method (methodological):** the literature might include some *qualitative* and some *quantitative* sources. You might group the resources by method and present them that way.

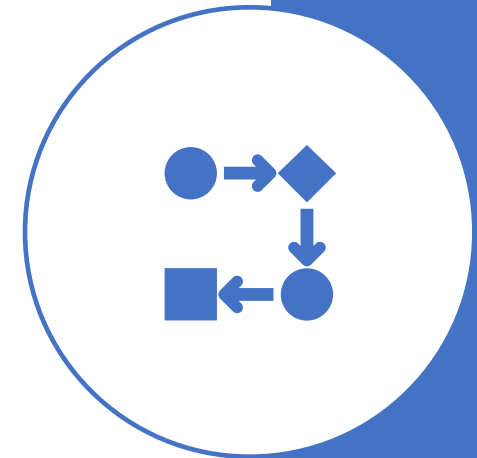


Literature review content

You won't just work through your sources one after another in your literature review. Instead, you will make *ideas* the backbone of the section. Consider addressing some of these questions and incorporate them into your literature synthesis table :

- What is the main argument each source makes?
- Why is the perspective significant?
- How do the sources contribute to the thesis you stated earlier?
- Where do the sources agree/disagree?
- How are the sources similar?
- Which sources make the greatest contributions to the topic?

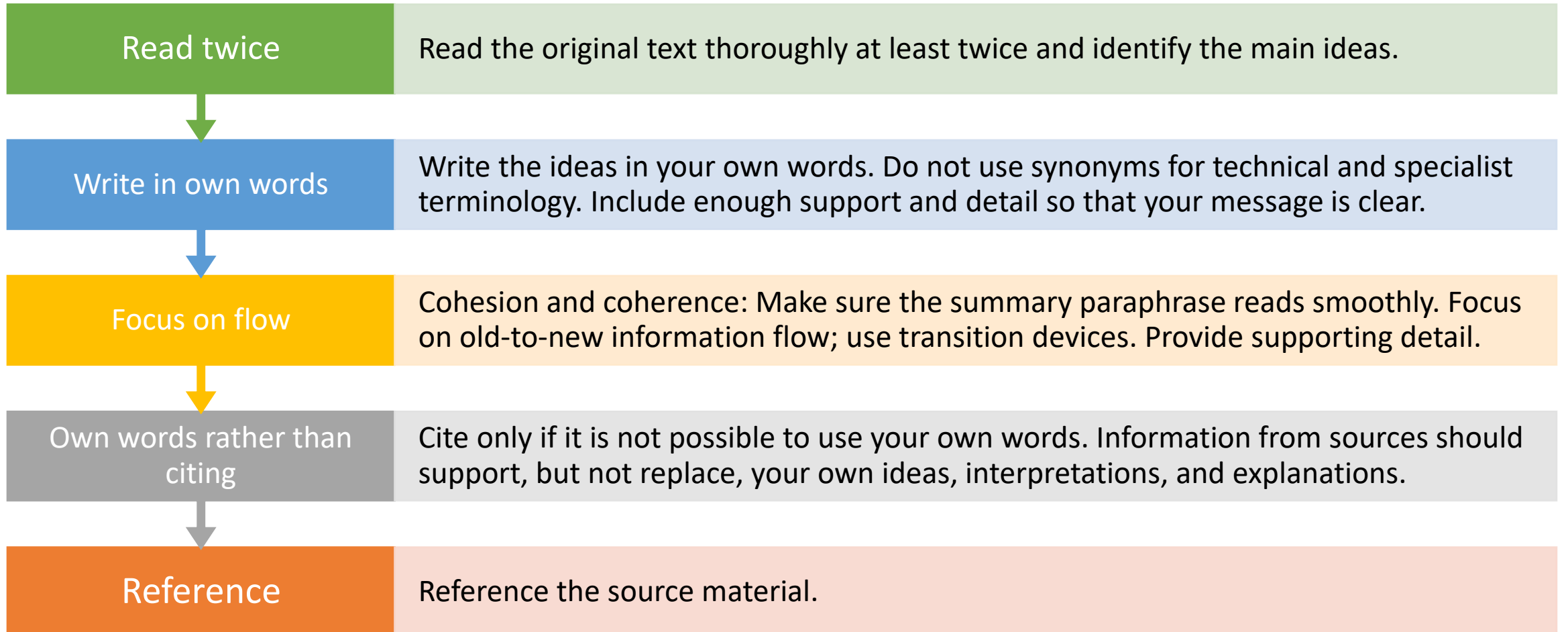
If you make use of **graphic organization** such as a **literature synthesis table** to keep track of and compare sources, it is easier to see where sources agree or disagree.



Integrating sources into the literature review

- Quote from sources sparingly and avoid long block quotations within a literature review, as they tend to drown out your own voice.
- When you do quote from the source, integrate it with your words like this:
DeKere (2010) supports this view, stating, “Including a registered dietitian in the treatment plan increases success rates by fifty percent” (p. 19).
- It is better to use **summary paraphrases** of other author’s words. (See the next section)
- Never refer to a secondary source – always go back to and cite the primary source. Don’t rely on someone else’s interpretation!
- Don’t trust trust generative AI (such as Chat GPT) here, as it tends to hallucinate (make up sources that don’t exist).

Paraphrasing steps





Paraphrasing techniques

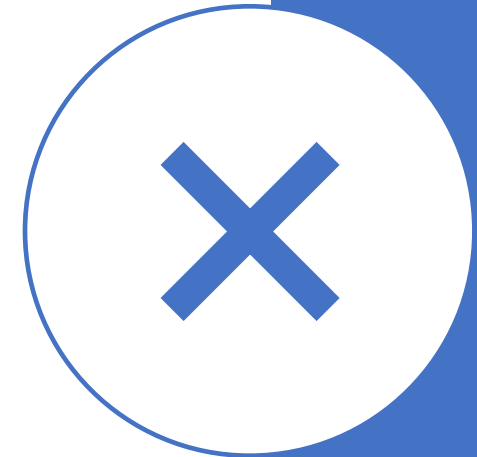
- **Change the vocabulary using synonyms**, for example:
seldom→*rarely*; *the 19th Century*→*the 1800s*;
procure→*purchase*; *saving*→*economizing*
- **Change the class of the word**, for example from verb to noun or adjective to verb, *such as in argument (n)*→*argue (v)*; *stimulating (adj)*→*stimulate*
- **Change the word order**, for example: 'The collapse of the health system caused a reduction in the quality of health services.' → 'Health service quality was negatively affected by the breakdown of the health system, which meant that children could not receive their routine immunisations.'
(Here the author added **implications**, as one might when writing a literature review.)
- Change between **active and passive voice**

All these techniques are used at the same time when one paraphrases. Make sure that your paraphrase accurately reflects the main ideas of the original text (Bailey 2015:82).

Plagiarism includes... (Scientific conduct at Swiss TPH and Uni Basel)

- turning in someone else's work as your own
- copying words or ideas from someone else without giving credit
- failing to put a citation (quotation) in quotation marks
- giving incorrect information about the source of a quotation
- changing words but copying the sentence structure of a source without giving credit
- copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not
- Using Artificial Intelligence without acknowledging it.

Familiarise yourself with the university's guidelines on citing AI tools that can be accessed here: [AI in learning and teaching | University of Basel](#)



Tense use in citations 1

Past tense, with the researcher activity as the agent

- Huang (2007) *investigated* the causes of airport delays.
- The causes of airport delays *were investigated* by Huang (2007). (Passive voice)

The past tense is mostly used to refer to single studies

Tense use in citations 2

Present perfect tense

- The causes of airport delays *have been widely investigated* (Hyon, 2004; Huang, 2007).
- Several researchers *have studied* the causes of airport delays (Hyon, 2004; Huang, 2007).
- There *have been* several investigations into the causes of airport delays (Hyon, 2004; Huang, 2007).

The present perfect tense is used to refer to areas of inquiry, or to **researcher activity over a period of time in the recent past**, or to **recent trends** in research.

Do not use this tense to refer to completed past activity that was completed. For that you use the simple past tense, e.g. 'In 1998 Donald et al identified various factors contributing to infection.'

Tense use in citations 3

Present tense

- The causes of airport delays are complex (Hyon, 2004; Huang, 2007).
- Airport delays appear to have a complex set of causes (Hyon, 2004; Huang, 2007).

The present tense refers to the current state of knowledge.

Reporting verbs 1

Consider, say, suggest, believe, think (often used with 'as'), for example:

- 'As Williams suggests, the first visit to the general practitioner is crucial.'

Argue, assert, claim, conclude, confirm, contend, insist, maintain, state, warn, for example:

- 'Williams maintains that xenophobic attitudes have had an influence on the increase in attacks on foreigners.'

Demonstrate, explain, indicate, note, prove, reveal, show, for example:

- 'As Williams demonstrated, cognitive behavioural therapy influences participants' decisions.'

Reporting verbs 2

Accept, agree, deny, doubt, for example:

- In her research on study habits, Smith doubts/questions Williams's findings on the positive influence of cognitive behavioural therapy.'

Blame, condemn, criticize (to take a critical or negative position), for example:

- 'Shelley criticises therapists for their lack of understanding of victims' traumatic experiences.'

Describe, characterize, define, evaluate, identify, portray, present, for example:

- 'In his studies in the field of anthropology, Nicholson portrays these tribes as peace-loving.'

Identifying the source: useful verbs

- According to Y (2011),
- A and B's paper (2011) discusses/lists...
- C (2010) states/claims/argues/maintains that..
- D (2009) suggests/asserts/hypothesizes/
states/concludes/pointed out that /investigates
...
- E (2013) proposed/found/identified/
demonstrated that...
-was first reported/proposed/identified by/
by...

Citation activity, esp. in theses

Consider some of the theses/articles you have collected.

- Highlight 5 references to sources.
- Write down at least 4 different verbs that are used to refer to sources (for example: (Rogers (2004) *concluded* that...)).
- Compare your list with those of the other students and make a list of useful verb phrases.
- Look at the suggestions on the slides and try to incorporate them into your writing.



Referencing styles

- Follow the referencing system prescribed by your department, eg. Harvard style, Chicago style, the APA (American Psychological Association) style, Vancouver style
- Be consistent
- Overview of the styles
<http://www.newcastle.edu.au/service/library/tutorials/newcat/writing/page20.html> and
http://www.chicagomanualofstyle.org/tools_citationguide.html

Paraphrasing and formal language exercises

- Study the examples of paraphrases at the end of the script and do the exercises.
- Do the formal language exercises too.
- Compare your answers to the possible answers uploaded on ADAM.



Assignment 3: Reviewed assignment and reading with note-taking

The use of Artificial Intelligence tools are NOT permitted for this assignment.

1. Paraphrase and summarize 3 paragraphs from epidemiological journal articles given in the script. The summary paraphrases should be about one third to half the length of the original paragraph, but contain the essential information found in the original text, stated in your own words.
2. Read 3 articles that will form part of your literature review. Record sources and make notes about the content. Make a literature synthesis table to record the sources (See example on the slides for lesson 3) OR Write a source list with 3 notes about each source. This part of the homework is not reviewed but must be included in the same document
3. Review the summary paraphrases written by your review partner. Follow the review process, keeping to the due dates and following the assignment naming conventions.

This is a reviewed assignment: follow the review process and naming conventions as explained. Due dates for **upload** on ADAM and list of review partners: see 'Assignment instructions' document on ADAM.

